Unify Learn Developer Handoff

Oct 27, 2026 · @Dibble

Read this first

Unify Learn gives about 3,000 students in LASU's Faculty of Engineering one place for their courses. Each course has 12 weeks of interactive notes and practice questions. Lecturers, Heads of Class and student contributors feed it; the Unify team runs it. The semester started on 26 October 2026 and runs 16 weeks (12 teaching, 4 exams). We are paid at milestones in Weeks 4, 8 and 12, so **build in the order of section 11**.

What every screen must feel like. A student opens the app, knows in 3 seconds what to do next, and does it in one tap. Everything is written like a friendly person, never like a system. Nothing important is hidden behind gestures. If a screen makes someone think, it's wrong.

The code already exists (repo `DIBBLS/unify`, `apps/web`, React and TypeScript, one web app for phone and laptop). This handoff is mostly a front-end redesign of existing routes plus a few new features. Where this doc and the old UI disagree, this doc wins. Where this doc and Figma disagree, ask Dibble.

Who's who:

- **Student:** reads notes, takes quizzes, revises, checks the timetable, marks attendance.
- **Head of Class (HOC):** a student with extra tools. Sets the class timetable, posts to the class, passes lecturer materials to Unify, scans attendance, confirms which lecturer teaches a course. There can be 1 to 3 per class.
- **Contributor:** a student who did well in a course and is paid to write its weekly notes and quizzes.
- **Lecturer:** sends materials, posts announcements, checks notes, sees class reports, opens attendance.
- **Unify team (admin):** checks contributors' work within 24 hours, assigns and reassigns, approves staff, sets up courses.
- **HOD and Dean:** department and Faculty panels. Due by Week 8; design being refin**ed**.

How to read the Figma file

File: Unify_Demo. Everything is on two pages.

Page "Online Class_Big D", section Unify_MoodBoard_2: phone screens. Rows are labelled in big green text, with a grey note under each label saying what the row is:

Row	What's in it	Status
SYSTEM (left column)	Colours, components, Box Boy Illustrations, Box Boy Avatars (round)	Final
ONBOARDING	Splash, slides, sign-up, department, courses, 7b goal, 7c study rhythm, 7d Pick your Box Boy, done	Final
HOME & CGPA	Home states, CGPA planner, Welcome back, Notifications, Loading	Final
COURSE & LEARNING	Course, outline, practice, note, quiz (MCQ, fill-in, maths keypad), results, revision, streak	Final
TIMETABLE & CLASS	Week and day view, lecturer check-in, Class tab, guide	Final
ME & SETTINGS	Me, edit profile, settings, notification settings, become class rep	Final
CLASS REP TOOLS	Post, edit timetable, contributors	Final
CONTRIBUTOR	Dashboard, compose, manual builder, quiz builder, review, live	Final (front-end redesign of existing code)
LECTURER	Onboarding, Home, Timetable, Course page, Review, Send materials, Post, Attendance list, Me, journey map, Demo lecturers	Demo built on assumptions; see section 7
Attendance row	Setup, Home with scan open, pass, HOC week, scanner, confirm a lecturer, My attendance	Add-on, not in the proposal; see section 8

Row	What's in it	Status
DEAN & HOD	Faculty, department, level, report, phone summaries	**Paused. Do not build.**
DARK MODE	Dark versions; "CHOSEN" = option A	**Paused.** Light is the only theme at launch
Push notifications	13 push messages with when they're sent	Final
STATES	Errors, empty screens, offline	Final

Page "Unify_MoodBoard_Desktop": laptop screens. Admin panel (6 screens) and the student app on desktop (Home, Course, Note, Quiz, Timetable). These are a first pass and "could use some work"; build the layout and behaviour, expect visual polish later.

Prototype. Press Play from these starting points to see the flows: Student app from splash, Contributor, Lecturer onboarding, Lecturer Home, Attendance student, Attendance Head of Class, Admin panel, Student desktop.

Sample data. Every name, number, course and date on the screens is sample data that shows the intended format. Real data comes from the API.

Design system

Colours: use the Figma variable collection "Unify · Roles". It names colours by job (page background, card, text, muted text, header, buttons, status chips), with Light and Dark values. Build the theme from these roles, not from hex codes, so dark mode can be switched on later without touching screens. The older collection "Unify" holds raw colours and is reference only.

Key colours: Unify green `#50ff90` (headers, small accents, active tab), black `#0a0a0a`, page off-white `#f5f5f0`, amber `#fbbf24` (the main action button on every screen), deep green `#15803d` / `#16a34a` for green text.

Type:

- **Bowlby One SC:** big headlines only, in capitals ("GOOD MORNING, JOSHUA.", page titles).
- **Inter:** everything else. Medium for body, Semi Bold for labels, Bold for titles and buttons.

- **Playfair Display Bold:** the "Unify Learn" wordmark only.
- **STIX Two Text Italic:** maths in notes and the maths keypad.

Shape and size:

- Cards have 20 to 24px corners.
- Bottom sheets have 40px top corners.
- Main buttons are 55px tall with 24px corners.
- Tap targets are at least 48px.
- Phone frames are 430 × 932.
- Laptop frames are 1440 wide, with a 248px black sidebar.

One main action per screen: the amber button. Secondary actions are black or outlined. Never two amber buttons on one screen.

Box Boy is the brand. Use only the images in the **Box Boy Illustrations** frame and the **Box Boy - Personas** frame; never older Box Boy art. Students choose one of 5 round Box Boy avatars (Box Boy Avatars · Round); there are no photo uploads for students, and no letter avatars anywhere. Lecturers may upload a photo, or keep a lecturer Box Boy (art coming: black box, white U, blazer). Push notifications use the Box Boy icon, not the logo.

Language. Plain, friendly, short. A little Nigerian warmth on student screens ("You don enter!"), calmer and respectful on lecturer screens. Never "carryover": say PRIORITY. Say "save", not "download". Lecturers are not all professors, so never address them as "Prof".

Lecturer screens: bigger text (17px body), high contrast, words on every button.

Student app, section by section

Onboarding

Intent: get a student from "what is this?" to their own courses on Home in under 2 minutes, and make it feel like theirs.

1. **Splash:** wordmark, "Own your academic journey", footer "Powered by Unify". That footer is a swappable slot for a future partner's name. No LASU branding on the splash.
2. **Slides 1 to 4,** then **sign-up:** Google, or email and password. "What should we call you?" asks for the **display name** used on the dashboard.

3. **Department:** School (a dropdown, not locked; LASU is the only option for now), Faculty (Engineering now; Environmental Sciences and Agriculture coming), Department, Level.
4. **Pick your courses:** first or second semester. Courses from other levels are allowed, and they show as PRIORITY.
5. **7b "What are you working towards?":** one of 5 goals. It sets where we start them: improve grades → a quick quiz; build a habit → one small weekly task; understand courses → a worked example first; study with others → teach-back prompts; balance school → short sessions.
6. **7c "When do you read best?" and focus length.** These set reminder times and the Pomodoro length.
7. **7d Pick your Box Boy:** 5 avatars, or "Surprise me". Changeable later in Me.
8. **Push permission, then "Bag that first class", then Home.**

Two names. The display name is for the dashboard. The **full name exactly as on the LASU ID card, plus matric number**, is collected separately in a short required 3-step setup (Settings, and prompted before the first attendance scan). It says "This is for attendance". The matric number never shows on the dashboard, only in Me.

Home

Intent: in 3 seconds, the student knows the one thing to do next.

- **First open:** a target slider with "Set my target", a CGPA preview, and the course list straight away. "+ Add course" is a small link, never a step.
- **Target set:** Box Boy holds the target sign, with an edit badge. A **hide numbers** option replaces the numbers for privacy (the "numbers hidden" state).
- **Cards, in order:** Continue (the next topic, black card, amber arrow); Today's revision; Next class ("Lecturer confirmed" or "Not confirmed"); My courses (PRIORITY chip for courses from other levels).
- **Attendance banner (amber):** shows at the top only while a scan is open for one of their classes. "Show my pass" opens the pass.
- **Class tab badge:** a 22px red circle with a bold count of unseen posts, never a slim pill.
- **Welcome back** replaces Home after 5 or more days away: what moved, with one main action.

CGPA planner

Intent: many students want a clear, honest path to 5.0.

1. **Results so far:** a grade per course, from which the CGPA is computed. A shortcut "I already know my CGPA" skips this step.
2. **Where do you want to finish?** Each option gets an honest label based on the units left: out of reach, very difficult, stretch or achievable. Then "What's driving this?" chips.
3. **The plan:** the GPA needed each semester, a roadmap, and a what-if showing the projected result and the highest possible result.

Analytics records the target band only, never real grades.

Course page

- **Week statuses:**
  - Done
  - In progress
  - **Locked:** opens when the previous week's quiz is passed, 6 out of 10.
  - **Not published:** shows "Notes land by <date>" with **Notify me**.
- One topic per class. Each week shows "Pending lecturer review" or "Reviewed by lecturer".
- Tabs: Weeks, Outline, Practice. Past questions in Practice stay **locked until the Faculty approves them** (copyright).
- **Save for offline** keeps a whole course on the phone. **Report a problem** sends the issue to that course's contributor.
- Students who register late see everything already posted and published.

Class notes

The design is Bowlby headings and Inter body. A note is built from these blocks:

- the learning goal box
- text cards
- formula cards (STIX)
- symbol tables
- worked examples with try-then-reveal steps
- quick checks

- a **Pulse check** of 3 questions at the end of each topic

Focus mode (pro feature, later): reading blocks the length chosen in onboarding (default 25 minutes), then a 5-minute break with pop-up recall questions and a teach-it-back prompt.

Quizzes

- **Format:** one MCQ per screen, plus theory questions with model solutions. One timer for the whole quiz, set by the contributor. Answers show only after submitting.
- **Copying and screenshots:** turn off text selection and the copy menu. A web app can't fully block screenshots, so the wording promises "discouraged", not "impossible".
- **Answer types:**
  - number answers use the number keypad
  - formula answers use the **maths keypad** (tabs: 123, f(x), Greek, Units), and answers are checked as **expressions**, so $8/27 \cdot \rho \cdot V^3$ and $0.296\rho V^3$ both count
  - **theory answers:** the app marks the final answer, then shows the model solution so the student can compare their working
- **Offline:** answers save on the phone and submit when the connection returns. The timer pauses while offline.
- **Results:** the score; passing (6 or more) opens the next week. Then answer review with explanations, and "Worth another look" links back to topics.

Revision and streaks

- **Spaced revision:** questions come back after 1, 3, 7, 14 and 30 days. At most **10 questions a day**.
- **Streak:** a day counts if today's revision is done or a class topic is finished. A day runs **4 AM to 4 AM**. Weekends never break a streak. A missed day restarts the streak quietly, with no shaming.

Timetable

- **Week view by default:** Monday to Friday, with a now-line. Changes are marked: moved, cancelled, test. A "Changes this week" list sits underneath. There's a toggle to Day view.
- **Who sets it:** the base comes from the Faculty timetable, and the HOC adjusts it. A lecturer's change wins.

- **Lecturer check-in:** "Is <lecturer> in class?" A class is confirmed when the lecturer taps Confirm, or when **3 classmates** confirm.

Class tab

Posts from lecturers and HOCs (test, assignment, exam, notice). Official posts are marked "From Dr. ___". It's announcement-only, with no chat. There's an empty state, and late registrants see the full history.

Me and Settings

Me has the Box Boy avatar, name, department and level, Edit profile, streak cards, Notifications (with a count), Guides, Saved for offline, class reminders, and Log out.

Settings (the gear) has:

- notification settings
- attendance setup
- "Are you your class's rep?" (request, then pending)

Guides (e.g. registering courses) are sample steps that the department must check before launch.

Head of Class tools

Intent: the HOC already does this work on WhatsApp and paper. Unify makes it faster, not more.

- The HOC is a student with a **"Class rep tools"** card in Me. It holds Scan attendance, Post to class, Edit timetable and Contributors. HOC screens are their own screens, not the student ones.
- **Becoming a HOC:** request it in Settings; the Unify team approves it (admin, People).
- **Edit timetable:** shows clashes and classes not confirmed yet.
- **Post to class:** the same post types as lecturers. The post is marked as coming from the HOC.
- **Contributors:** shows who writes each course and whether they're behind, plus **Upload notes from your lecturer**, which passes materials to Unify.
- **Confirm a lecturer:** when a lecturer adds a course this class takes, the HOC gets "Is this your lecturer?" with **Yes / No, not our lecturer**. Until they answer, the lecturer sees "Waiting for Head of Class". If two lecturers claim one course, the HOC can confirm

both (co-teaching).

- **Take attendance:** a button on each course week, whether or not the notes are out. It's **only active after the lecturer opens a scan**; a HOC can never start attendance alone.

Contributor side

Intent: paid students turn lecturer materials into great weekly notes, on time, and fix what students report. This already exists in code (`dashboard` , `studio` routes); **redesign the front end only.**

- **Navigation:** contributors use the same app with 3 tabs: Home, Studio, Me.
- **Home:** what's due this week, days left on each **7-day deadline**, materials received, student reports to fix, weeks live.
- **Studio flow: Compose → Review → Live.**
  - Compose offers AI Generate, Manual (the block picker) and External AI.
  - Notes use the same blocks students see.
  - The quiz builder takes objective questions and theory questions with model solutions, the timer and the pass mark.
- **Cap:** at most **2 courses per level** per contributor. The server enforces it; admin's cap audit lists anyone over.
- **Lifecycle of a week:**
  1. Materials arrive.
  2. The contributor writes the week.
  3. Submit.
  4. **The Unify team checks it within 24 hours.**
  5. It's published as "Pending lecturer review".
  6. The lecturer marks it reviewed.
- **Late:** more than 3 days late lets admin **reassign** the week.
- **Student reports:** each "Report a problem" goes to the contributor. They mark it Fixed or Not a problem, and the student is told.
- **Joining:** a recruitment graphic, then a Google Form (result in the course plus a sample topic). Admin imports and accepts applications.
- **Pay:** per approved week of notes plus its quiz, each payment cycle.

Lecturer side

Intent: a lighter load for the lecturer. Send a file once, post once, and every class taking the course gets it. This is its own product, not a tool that exists for students. **It's built on our assumptions; no lecturer has been interviewed yet.** Build what the proposal promises first (marked ★); keep the rest simple and easy to change.

Hierarchy rule (applies everywhere): always Faculty, then level, then department, then course. Show a level pill ("300 LEVEL") beside every course. Filters say "All departments", never "Everyone". One lecturer can teach a course shared by a whole Faculty level (e.g. 742 students across 6 departments) and also a single class (e.g. 400L ECE), and can teach 100L to 500L.

Screen	What it does	Rules
Onboarding (6 steps)	Welcome "SEND IT ONCE", sign-up, home department, search courses, let students know you, done	Sign-in with Google, email or phone; email preferred. Search covers any department and level, with a semester switch. Photo, bio, email and handles are optional.
Home	Today card, notes to check, profile prompt, courses by level	**One main button at a time:** before class, Confirm class (with a "Running late or cancelling?" link); from class time, Open attendance scan. Students see "Lecturer confirmed". 7 AM reminder push. No settings on Home.
Timetable	Class timetables, plus My own	Class timetables are set by HOCs; a lecturer's change wins. "My own" is private (office hours, duties).
Course page	Weeks, Report, Attendance tabs, with an All departments / ECE / MEE / CHE switch	Co-teaching: up to 3 lecturers per course ("with Engr. T. Balogun").
★ Send materials	Upload a photo or file for a course and week	Goes to Unify, not straight to students. Optional toggle: also share the file with students. The HOC route still works. Notes publish within 7 days.

Screen	What it does	Rules
★ Post to students	Step 1 course, step 2 departments (all, or tick some), step 3 type and message	Announcement-only, like a locked WhatsApp group. Students who register later still see it. Confirm before sending to hundreds.
★ Review a week	The note as students see it, with Mark as reviewed or Needs changes plus a comment	Reached from Home ("notes to check") and the course Weeks tab. Unpaid this semester, paid from next session.
★ Report	On Unify vs registered, notes opened, practice finished, quiz average, most-missed questions, Add practice questions	**Counts only, never student names.** It can split by department and level, and it prints.
Attendance	Sessions (e.g. Mon and Wed, Weeks 1 to 12) and each student's percentage	Wording is always "Came to 1 of 4 classes held". Rows show level, department and matric number, because names repeat. Save as PDF or Excel.
Me	Profile, my courses, "Waiting for Head of Class", sign-in, notifications, guides	Lecturers get their own mascot later.

Course confirmation: a course works as soon as the lecturer adds it, but it shows "Waiting for Head of Class" until that class's HOC confirms it.

Attendance

Intent: replace 35+ minutes of name-calling or paper lists in classes of up to 750, with a scan at the door that takes about 5 minutes and that nobody can sign for someone else. **Not in the Faculty proposal; it's an add-on.** Build it after the Week 8 deliverables unless Dibble says otherwise.

The flow:

1. The lecturer taps **Open attendance scan** (Home, at class time). HOCs of every department in that class get a push. Only the lecturer can open a scan.

2. Students see the amber "Attendance is open" banner on Home and open **their pass**. It's one pass per student, valid for all their courses.
3. At the doors, HOCs (1 to 3 per department) and/or the lecturer scan passes as students leave. Any scanner can scan any student in that course.
4. The scanner shows the student's Box Boy, name, level and department with a tick, or one of: "Already scanned", "Not registered for <course>", "Pass needs a refresh".
5. The lecturer taps **Close scan**. The list shows present, typed in and flagged, and who scanned each person, plus a **Spot check** that picks 10 random names to call out.
6. Each student sees their own percentage per course in My attendance. Below 75% they see a gentle warning to check their department's rule.

Security requirements: these must hold offline, on both phones.

Need	How
The pass is real	Once a day, while the student has data, the server issues a **signed pass**: student ID, ID-card name, matric number, department, level, registered courses, valid-until, and the student's device public key. It's signed with Unify's private key (Ed25519 or ECDSA P-256). Scanners hold only Unify's **public** key, so no scanner can forge a pass.
Only this phone can use the pass	At first login the phone creates a key pair with Web Crypto (ECDSA P-256, **non-extractable**, kept in IndexedDB). The public key goes into the pass. One active device per student; signing in on a new phone replaces the old key.
The code is fresh	Every 30 seconds the QR shows the pass plus the current time, **signed by the device key**. Scanners accept codes within about 2 minutes of their own clock; otherwise the student is typed in manually. A screenshot dies within that window. Keep the payload compact (binary, base64url) so the QR stays easy to scan.

Need	How
The scan counts once, for the right class	The scanner is opened for one session (course and date). It checks the course is in the pass, blocks duplicates on that phone, and stores the scan with who scanned it and when. Scans queue offline and upload later; the server checks every signature again, keeps the first scan and flags the rest.
Dead phone or no pass	The HOC types the matric number. The entry is marked **typed in** and shown to the lecturer.

Known limits (the people side covers them): no offline method proves someone is physically in the room. A student could relay a live code to a friend on a video call while the scan is open. The deterrents are the HOC at the door, opening the scan at an unannounced time, the spot check, and "scanned by" on every row. Students at the window are the lecturer's call. There are no student photos (Box Boy avatars only), so the scanner shows the name and avatar.

Admin panel for the Unify team

Intent: the team opens the panel each morning and sees exactly what needs a person today. It's a redesign of the existing `admin.tsx`, which is one page of folding sections, into a laptop panel with a sidebar. **Keep every existing function and API call; regroup them.**

New screen	Holds these existing modules	New behaviour
Overview	Analytics (stats, by role, sign-up and notes trends)	Proposal targets with on-track or behind; "Needs you today" list linking to each screen
Content pipeline	All content (`/admin/content`)	A board per week of notes: Materials received → Contributor writing → Submitted (24-hour check) → Live, lecturer to check → Reviewed. Late cards (more than 3 days) get **Reassign**; unassigned materials get **Assign**. Filters by department and "Late only".

New screen	Holds these existing modules	New behaviour
Contributors	Part of Users & roles (assigned courses)	Google Form applications (import, accept, decline); the payment cycle and export; a contributor table with weeks done, on-time rate and late weeks
People	Staff requests, Users & roles	Request cards with Approve and Decline; user table with invite by email, role filter and Change role. Removing a user always confirms first.
Setup	Universities, Courses per level, Academic session	Faculties per university with **Turn on** (Environmental Sciences and Agriculture are next). The semester switch. "Promote all students one level" in a red danger zone with a confirm step. Add and delete courses (delete confirms: all its weeks go).
System	Announce, App errors, AI models, Cap audit	Announcements to everyone or by role, with a confirm step and a recipient count. Errors list (Sentry status). Model health with Reset. Cap audit with "Trim to the earliest 2".

Links to **All content** and **Studio** stay in the sidebar.

Notifications, states, offline, tracking and privacy

Push notifications (the board in Figma has all 13, with their triggers):

- Written like a person, with one action each. Tapping one opens the exact screen it's about.
- **At most 3 a day per student**, except class changes (cancelled, moved, test), which always go out.
- **Quiet from 10 PM to 6 AM**, unless the student chose "Night owl".

- The morning brief comes at 7 AM or the student's chosen time, with **one** call to action.
- Lecturers get a 7 AM "Confirm today's classes". HOCs get "attendance opened" and "is this your lecturer?".
- The icon is Box Boy.
- The in-app Notifications screen (reached from Me) lists the same messages.

States (the STATES row): every error says what happened and what to do next, never a code:

- something went wrong (with "open saved notes")
- wrong verification code (tries left, resend timer, use email instead)
- course search with no results
- pass needs a refresh
- quiz saved offline
- lecturer with no confirmed courses yet
- loading a note
- the Class tab with nothing posted
- no classes today
- offline

Offline is normal, not an error. Saved courses, notes, the pass and quiz answers all work without data. Anything done offline queues and syncs later, including analytics events. Show "Saved offline" chips, never error banners, for things that work.

Tracking: follow the **Unify Learn Tracking Plan** doc. Add the sign-up funnel events before launch, and the rest in the first two weeks. One tool for the app and the website; PostHog is suggested.

Privacy (Nigeria Data Protection Act 2023):

- Analytics never stores names, matric numbers, phone numbers, emails or grades. People are random IDs plus role, faculty, department and level.
- Matric numbers are visible only to the student, their HOCs and their lecturer's attendance list.
- Lecturer reports, HOD panels and Dean panels show **counts only**.
- The CGPA planner keeps results on the student's account and never shares them.

Build order

The order follows what the Faculty pays for. Semester Week 1 started on 26 October 2026.

1. **Now to the end of Week 3 (15 Nov): students can sign up and learn.** Onboarding (including 7b to 7d and the ID-card name and matric setup), Home, CGPA planner, course page, notes, quiz (MCQ, number keypad, offline save), Class tab, timetable, Me and Settings, HOC tools, push notifications, states, and the sign-up tracking events. Target: 2,400 active students.
2. **By the end of Week 4 (22 Nov): content runs on time.** Contributor front-end redesign, the admin panel (pipeline, contributors, people, setup), and departments configured. Targets: 52 contributors submitting, notes published within 7 days, first Faculty report.
3. **By the end of Week 8 (20 Dec): lecturer side.** Lecturer onboarding, Home, Send materials, Post to students, Review a week and the engagement report (★ items in section 7). Then revision and streaks, the maths keypad and the desktop layouts. Target: 48 of 60 lecturers set up.
4. **After Week 8:** attendance (section 8), focus mode and other pro features, and lecturer Box Boy mascots.

Paused, do not build yet: dark mode (option A is the starting point when it resumes), online-class attendance, and chat between lecturers, HODs and the Dean.

HOD and Dean panels, due by Week 8: the proposal's Week 8 milestone (₦3,000,000) lists "lecturer side, lecturer engagement reports **and both panels**". The HOD and Dean panels are part of the Week 8 delivery. Dibble is still refining their design; build them from the DEAN & HOD row once he signs them off, in time for Week 8.

Open questions and links

Still open:

- The exact name of the Industrial department (we use "Industrial & Production"); confirm all 6 department names.
- Whether attendance passes last 1 day or up to 7 days between refreshes.
- How many devices a student may switch between per semester.
- Payment rates for contributors, and for lecturer review from next session.
- When past questions in Practice can be unlocked (needs Faculty approval).

Related docs:

- Lecturer Plan: lecturer decisions, attendance design, Dean and HOD panels.
- Contributor Plan: the weekly loop, decisions, admin panel.
- Unify Learn Tracking Plan: every event and funnel.
- Student App Plan and Onboarding Plan: in Dibble's docs.
- Figma: Unify_Demo.

Questions go to Dibble. When the design and this doc disagree, ask before building.